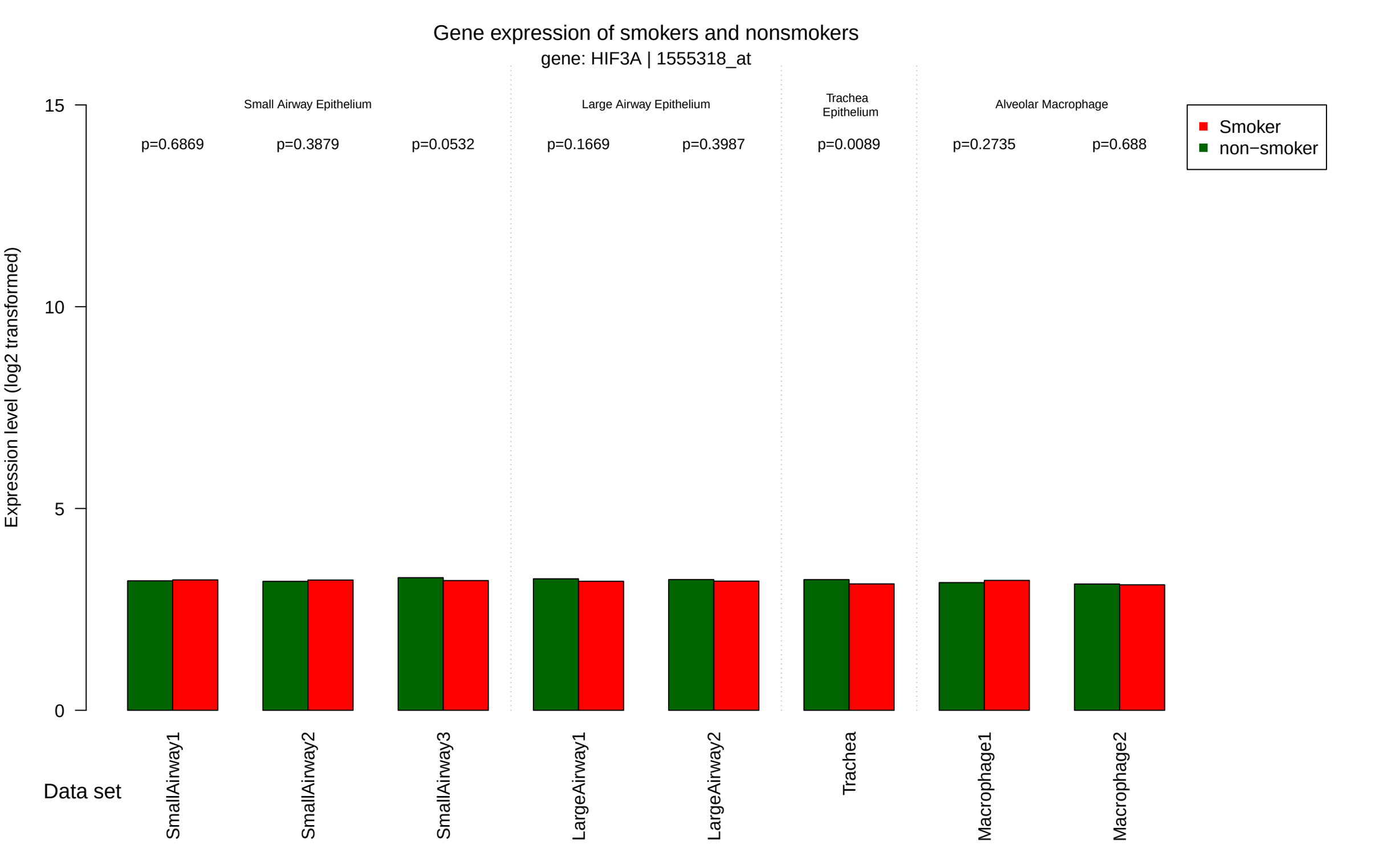
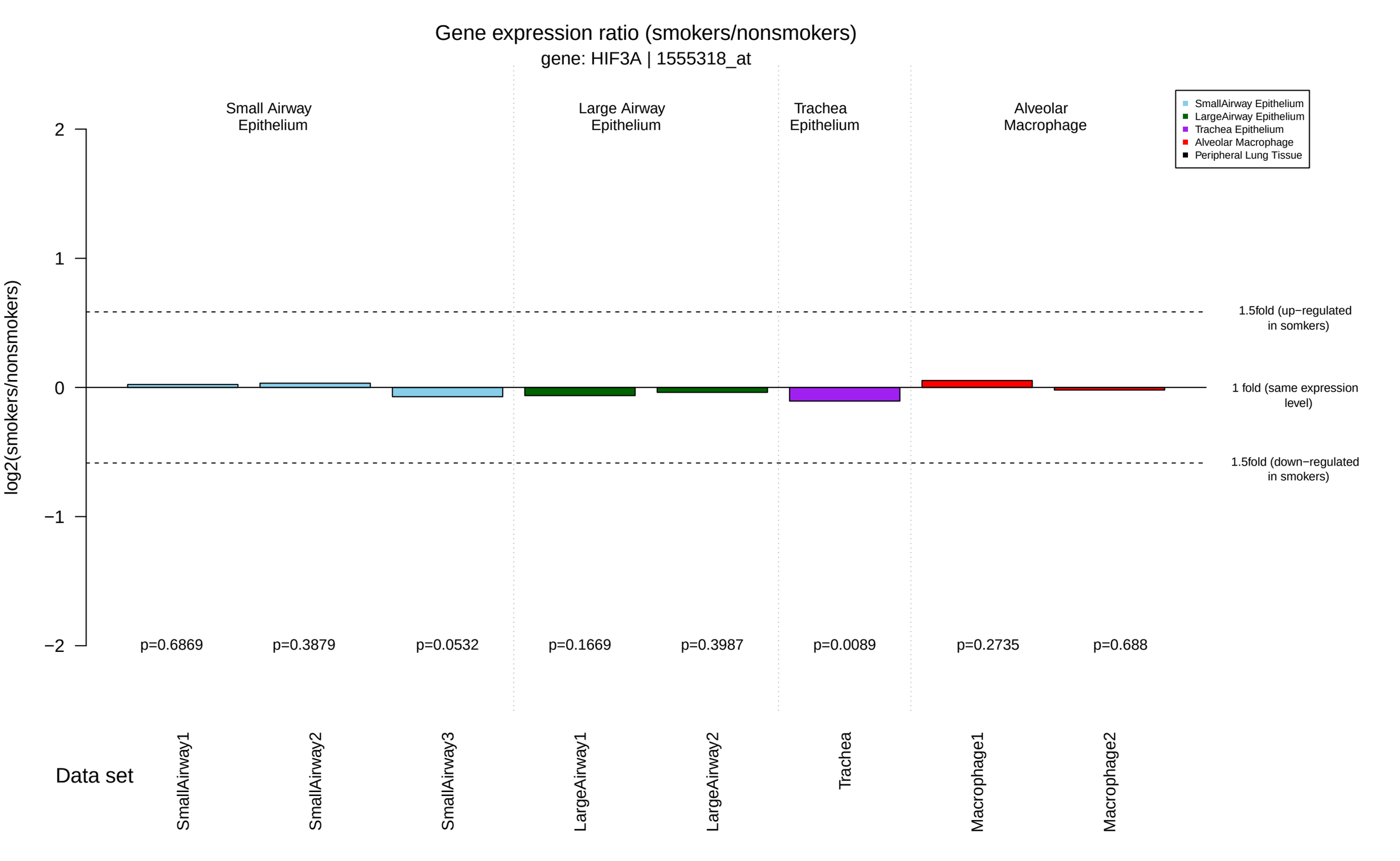
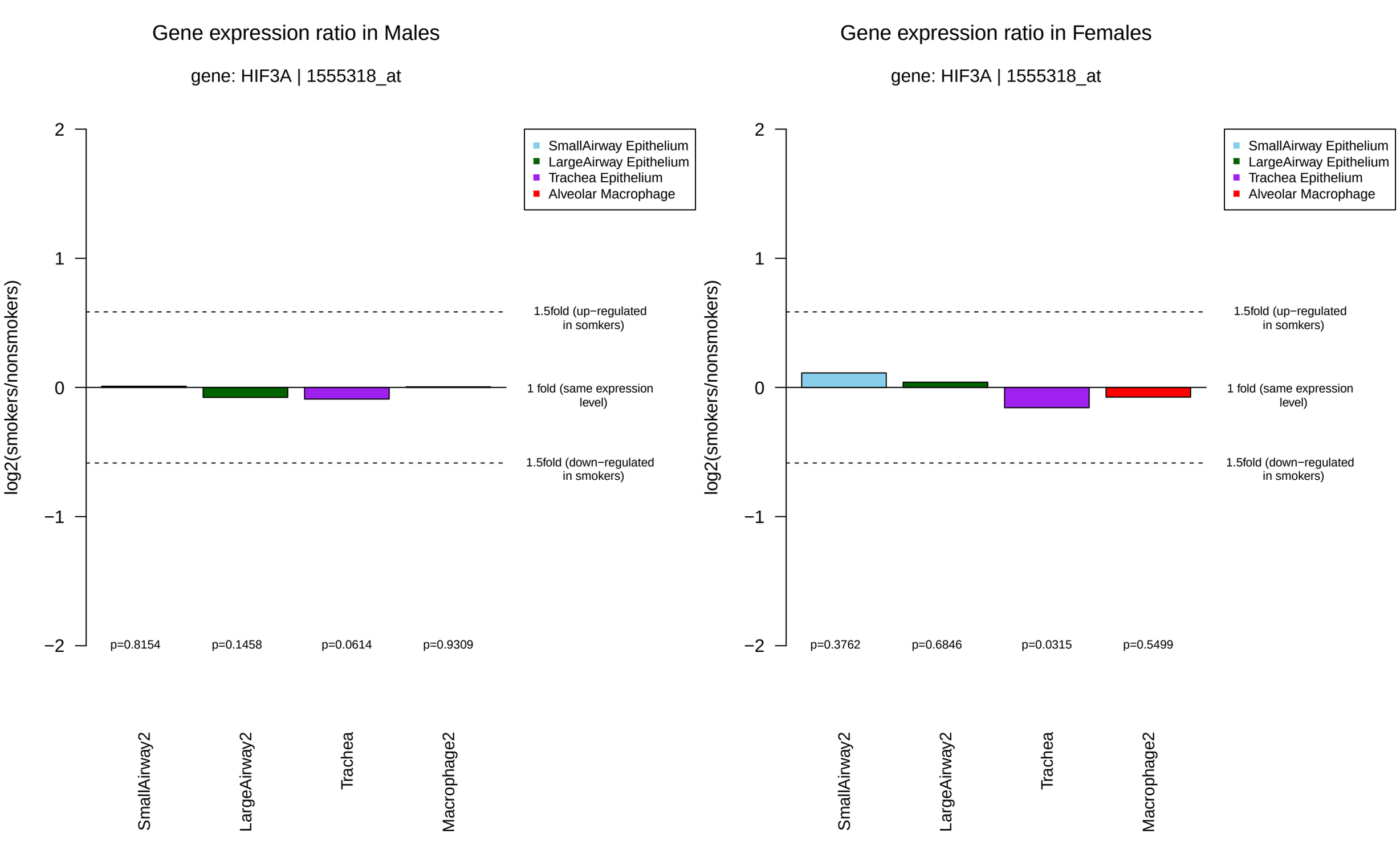
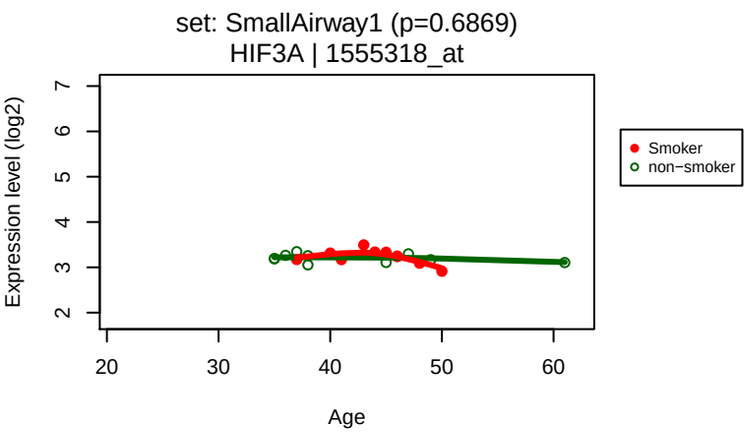
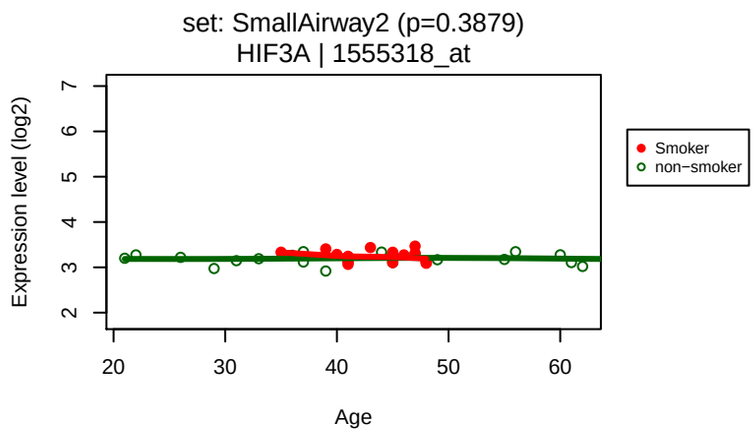
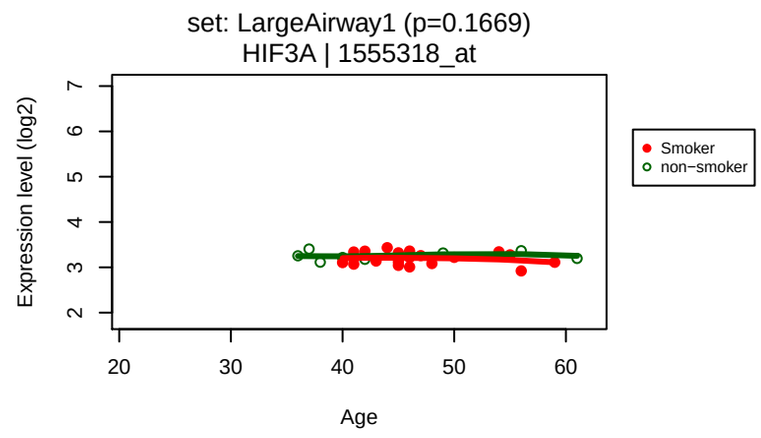

Gene expression of smokers and nonsmokers
gene: HIF3A | 1555318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6869
p=0.3879
p=0.0532
p=0.1669
p=0.3987
p=0.0089
p=0.2735
p=0.688
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

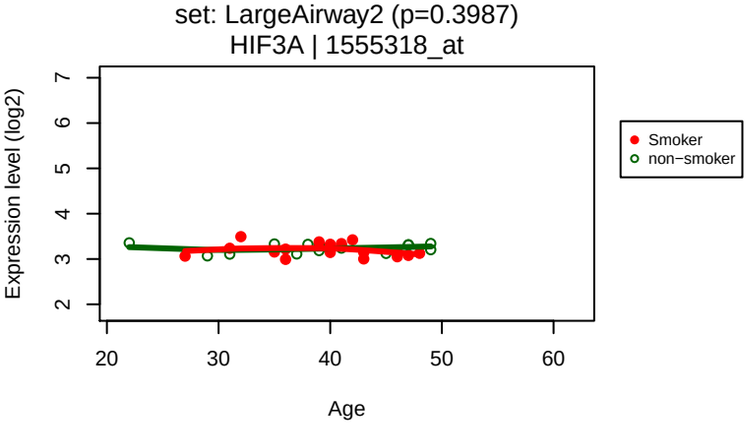

Gene expression ratio (smokers/nonsmokers)
gene: HIF3A | 1555318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6869
p=0.3879
p=0.0532
p=0.1669
p=0.3987
p=0.0089
p=0.2735
p=0.688
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

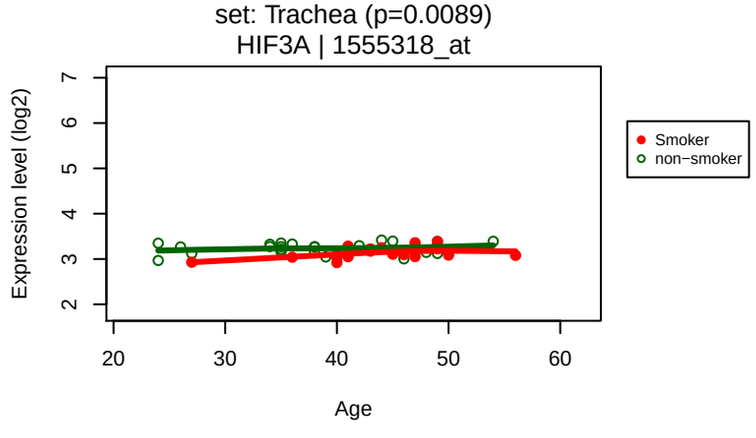

Gene expression ratio in Males
gene: HIF3A | 1555318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8154
p=0.1458
p=0.0614
p=0.9309
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIF3A | 1555318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3762
p=0.6846
p=0.0315
p=0.5499
SmallAirway2
LargeAirway2
Trachea
Macrophage2

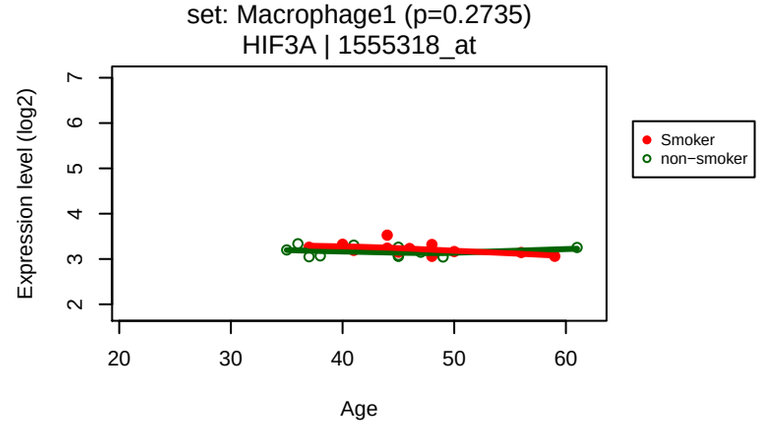

set: SmallAirway1 (p=0.6869)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker

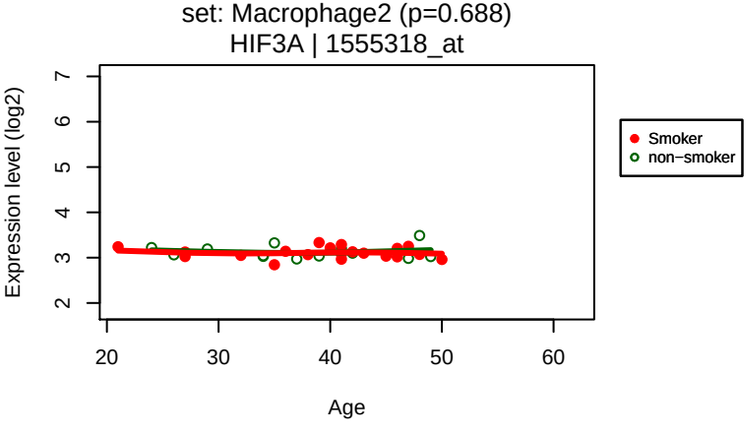

set: SmallAirway2 (p=0.3879)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker

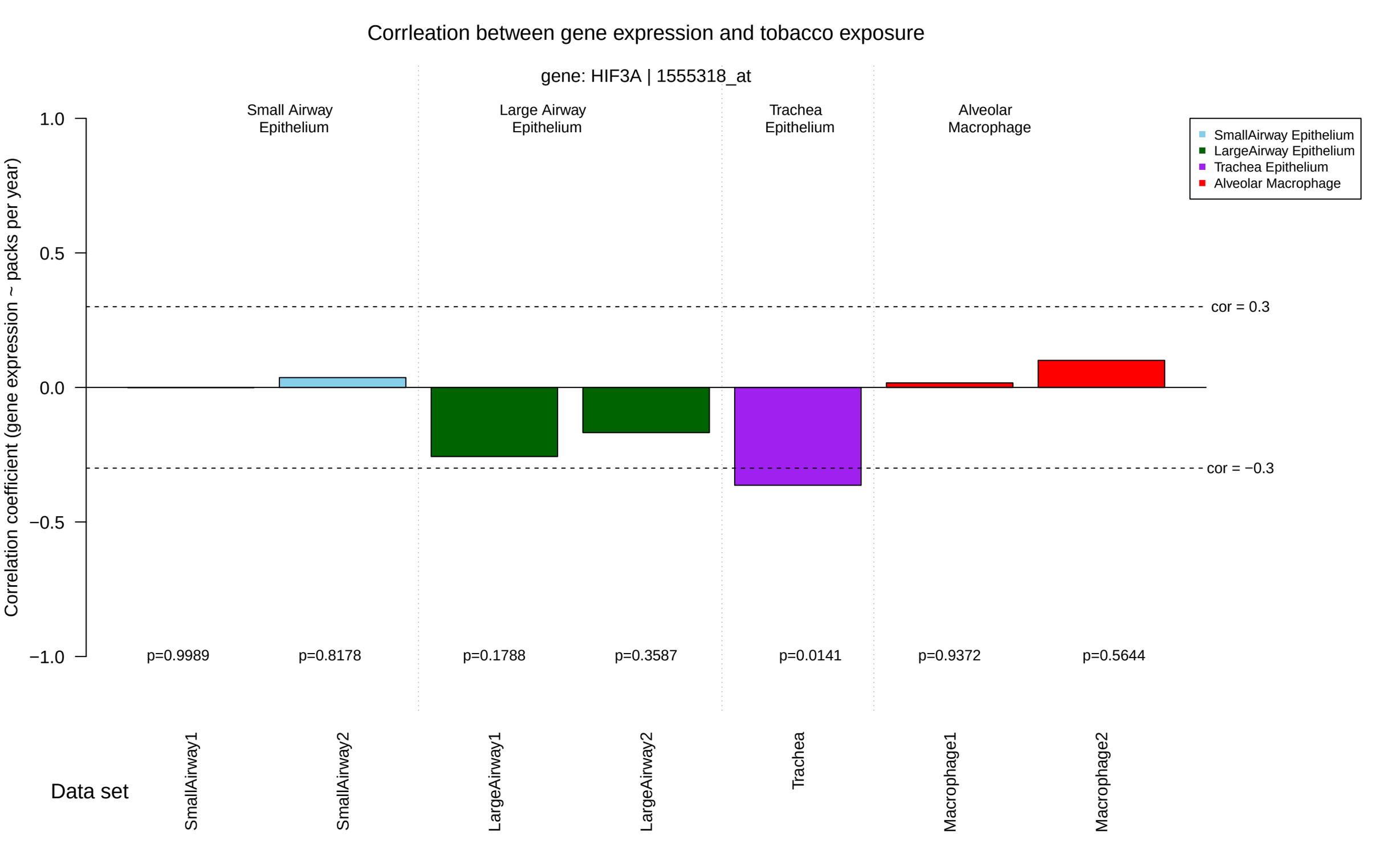

set: LargeAirway1 (p=0.1669)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3987)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0089)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2735)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.688)
HIF3A | 1555318_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIF3A | 1555318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9989
p=0.8178
p=0.1788
p=0.3587
p=0.0141
p=0.9372
p=0.5644
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2